

ECOM30001 / ECOM90001 Basic Econometrics

Tutorial 2

Zheng Fan

The University of Melbourne

Housekeeping

Zheng Fan

- Ph.D in Economics
- Contact: Canvas mail box

If you need any help, don't be shy:

- Weekly consultation
- Ed discussion board
- Consult Stop 1 for special consideration

Before posting questions, make sure you have gone through the materials on Canvas and existing posts on Ed discussion board!

Summary Statistics & Intercept-Only Models

1. Subsetting in R:

- Full sample: `tut2 <- import("houses.csv")`
- Pool subset: `subset(tut2, pool == 1)`
- No-pool subset: `subset(tut2, pool == 0)`

2. Intercept-Only Regression:

$$Y_i = \beta_0 + u_i \implies \text{R: } \text{lm}(\text{price} \sim 1, \text{data} = \text{tut2})$$

Key Result

The estimated intercept $\hat{\beta}_0$ is ****always equal to the sample mean**** $\bar{Y}$ of the dependent variable for that sample/subset!

Derivation: OLS Estimator in Intercept-Only Model

Model: $Y_i = \beta_0 + u_i$. Minimize **Sum of Squared Residuals** (SSR):

$$S(\beta_0) = \sum_{i=1}^n e_i^2 = \sum_{i=1}^n (Y_i - \beta_0)^2$$

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Step 1: First-Order Condition (FOC)

$$\frac{dS(\beta_0)}{d\beta_0} = -2 \sum_{i=1}^n (Y_i - \hat{\beta}_0) = 0 \implies \sum_{i=1}^n (Y_i - \hat{\beta}_0) = 0$$

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Step 2: Expand Summation & Solve for $\hat{\beta}_0$

$$\sum_{i=1}^n Y_i - \sum_{i=1}^n \hat{\beta}_0 = 0 \implies \sum_{i=1}^n Y_i - n\hat{\beta}_0 = 0$$

$$n\hat{\beta}_0 = \sum_{i=1}^n Y_i \implies \hat{\beta}_0 = \frac{1}{n} \sum_{i=1}^n Y_i = \bar{Y}$$

Dummy Variable Regression

Regressing house price on the pool dummy variable:

$$\text{price}_i = \beta_0 + \beta_1 \cdot \text{pool}_i + u_i$$

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- **Slope coefficient $\hat{\beta}_1$:** Difference in sample means:

$$\hat{\beta}_1 = \bar{Y}_{\text{pool}} - \bar{Y}_{\text{no pool}}$$

Wrap up

Thanks for your attention!

Let me know if you have any questions or concerns.